iGEM IISER Kolkata 2026

Inclusivity · Project FINS

Interview Notes

Interviews with IISER Kolkata faculty and administrators on women in STEM, and with the fisherfolk of the Ganga and a fish ecologist

---

Twelve interview records, reproduced as written.

Contents

Part A · Women and other minority communities in STEM at IISER Kolkata

Interview 1: Dr. Rahul Banerjee, Dean of Faculty Affairs

Interview 2: Dr. Rangeet Bhattacharyya, Dean of Academic Affairs

Interview 3: Dr. Anindita Bhadra (DBS)

Interview 4: Dr. Malancha Ta (DBS)

Interview 5: Dr. Anuradha Bhat

Interview 6: Dr. Neelanjana Sengupta, ADoFA (DBS)

Interview 7: Dr. Koel Das, HOD, Mathematics and Statistics

Interview 8: Dr. Rumi De (DPS)

Interview 9: Dr. Rumi De, follow-up to Interview 8: local NGOs and skills training

Part B · Fisherfolk and fish ecology

Interview 10: Mohadep Dutta, fisherman (Part 1)

Interview 11: Gobinda Shikdar and Sukuma Sarkar, fishermen (Part 2)

Interview 12: Dr. Anuradha Bhat, fish ecology

Part A: Women and other minority communities in STEM at IISER Kolkata

Interview 1: Dean of Faculty Affairs (DoFA)

Faculty recruitment, women in hiring, and hostels

Interviewee: Dr. Rahul Banerjee, Dean of Faculty Affairs (DoFA), IISER Kolkata

Why we went

- While we were compiling the faculty data of all the IISERs, we found some ratios that were difficult to understand. The ratios we found needed an explanation, and we sought explanation from trustworthy sources such as the deans.
- We asked how faculty are recruited, whether the government sets any target for women, why women apply less, why IISER Kolkata's share of women faculty is lower than at TVM and Pune, why women are fewer in Physics and Maths than in Biology and Chemistry, and whether the institute does anything special to attract women applicants.
- We were also curious about the exact number of faculty from the reserved categories at IISER Kolkata (since the data was not publicly available).

What we got

1. How faculty are recruited

- **The institute first follows the norms set by the Government of India**. This covers not only the male to female ratio but also caste and all other social representation.
- Each department then states its needs. These are mainly teaching needs: which new courses are being offered, who will teach them, and whether they fit national priorities and prepare students for research or entrepreneurship. The advertisement goes out on the basis of this feedback.
- Shortlisting goes through several committees. The Departmental Faculty Advisory Committee (DFAC), which he chairs as DoFA, shortlists with help from department representatives. It then goes to the Institutional Faculty Advisory Committee (IFAC), chaired by the director, which can send it back.
- The final interview has external experts, with the director as chair. Apart from the DoFA and the department chair, no departmental representatives are present. It takes days of questioning.
- The DoFA office, with an officiating deputy registrar, checks eligibility and due diligence at the back end. The whole process takes about a year, and the institute is trying to bring it down to 4 to 6 months, preferably 4, as per government norms.

2. Policy on women in hiring

- There is no government guideline on what percentage of faculty should be women, and no reservation for women. He said we are right that there is no such government rule.
- The institute has set its own target. In the latest special joint recruitment, it offered 8 candidates, of whom 4 were women (50%), and all joined. [Later, one left, making the numbers 3 women and 4 men.]
- From 2013 to last year there were more than 40 recruitments, of which only 8 were women, roughly 20%. The institute is now trying to raise this to 50% of recruits so that the overall ratio improves.

- He added that the institute does not want to hire anyone who cannot teach properly, so teaching quality comes first.

3. Few women apply

- In some departments and specialisations the institute receives one or no applications from women. In Chemistry, for example, close to 200 applications came in but hardly 20 were from women. He said if there were around 50, it would have helped a lot.
- He said he is a bit surprised and unhappy, since the institute is keen to hire women. He thinks there is a social-background issue and wants young women like us to aspire to become faculty.
- He tells students in class about Dorothy Hodgkin, who let her women students bring their children to her office and looked after them while they worked in the lab. His point was that women set the example for one another.
- He guessed that women may feel they will not get a fair chance in a patriarchal society, but said he does not know. In reality, the institute is looking for strong, motivated women scientists.

4. Why IISER Kolkata is at 14% while TVM and Pune are at 22 to 24%

- We shared that our share of women faculty has stayed at about 14% since 2019. He said his hunch is location, including the safety of Kolkata, and spouses finding jobs elsewhere.
- He recently lost a woman candidate who said she would join only if her husband was offered a job. The institute could not do that because appointments must follow a process.
- Earlier, a faculty member left for Pune because her spouse works there, and others have preferred places closer to a spouse's job. He said this is the problem IISER Kolkata is suffering from at the moment, though things have improved over the years.
- The institute is trying as many ways as it can to attract more women candidates, and said more effort is required.

5. Biology and Chemistry versus Physics and Maths

- He said he does not know the reason.
- In the current round they did not get many Physics candidates. They could hire one woman in Computer Science, but none in Physics. Physics has only one woman at the moment, and he hopes this changes.
- He suggested we talk to the women faculty in Physics and Maths, that is, Dr. Koel Das and Dr. Rumi De, and ask why women are not joining their departments as that would be more insightful.

6. Why hiring is slow, and promotions

- The institute has about 130 faculty, of whom about 87 are professors, so over 60% are already full professors. Promotion happens with time and performance, with checks and balances. At the first promotion they try to encourage people. From this year, student feedback is part of the promotion criteria.

- Strong applications are arriving, but mostly in areas the institute already covers, such as stem cell research or porous materials. It needs people in new areas such as disease biology, bioinformatics, optogenetics, and AI and ML in biology. Selection is therefore sluggish.
- He said that without good faculty, students will be frustrated and will not join the institute.

7. Advertising and special measures

- Advertisements are put on all social media, such as Twitter and Facebook, and on the institute website. He did not describe any women-specific channels.
- Since there is no reservation for women, the institute creates its own: for example, if it hires ten people, it tries its best to have more women among them to balance the ratio. It has been trying for the last five years.
- He said more women candidates are slowly coming, but it is still sad that the institute does not get enough.

8. Hostels for women

- He said he is not happy that the female PhD students are staying outside campus, since the area is not developed.
- The next target is to build a hostel as soon as possible. The plan is a four-floor hostel, with the first floor opened first so that girls can move in before the boys.
- His reasoning: undergraduates will see senior women students on campus, get an idea of which field to choose, and then choose IISER Kolkata. Women staying outside gives a bad first impression.

9. Regarding faculty from SC, ST and OBC

- We enquired about the very low number of faculty from SC, ST, and OBC categories at IISER Kolkata.
- He informed us that prior to 2019, although the government used to recommend keeping 7.5% ST, 15% SC, and 27% OBC candidates, there was no mandate as such. The mandate was implemented in 2019, wherein new faculty hirings would be stalled unless the required quota of reserved seats was filled in the roster.
- Therefore the recruitment has increased post-2019 in IISERs, IISc, and all other institutes, including IISER Kolkata. Hence IISER Kolkata is not an isolated case but a general trend that has been observed across the country.
- The initial low numbers of SC, ST, and OBC faculty members can be explained by the already deeply rooted social issues, which have been researched well in many other case studies.

10. What gaps still remain?

- We couldn't look at the data of how many women applied as compared to how many were offered a position, or how many joined. This was because the data was not stored. Thus, we cannot comment on the exact quantification of the same. However, as per the accounts we've heard, it's a general trend that women apply less than men for the same vacancies.
- We couldn't find the domicile data of the faculty members.

- The data of how many SC, ST and OBC candidates applied as compared to how many were offered a position, or how many joined was not available. Hence, we could not proceed further in this regard. We recommend the next teams to work on finding this data.

What we did next

- We interviewed the women faculty at IISER Kolkata in an attempt to understand the issue. This included Dr. Koel Das (HOD, DMS; the only woman faculty member in DMS), Dr. Rumi De (DPS, the only woman faculty member in DPS), Dr. Neelanjana Sengupta, Dr. Malancha Ta, Dr. Anindita Bhadra and Dr. Anuradha Bhat. We also talked to Dr. Rangeet Bhattacharyya in an attempt to understand the trends.

Interview 2: DoAA

Admission and gender data by department, and why women choose different majors

Dr. Rangeet Bhattacharyya, Dean of Academic Affairs (DoAA)

Why we went

- To verify whether our data was correct. He helped us verify it.
- To ask why he thinks the share of women is so low. (He said this is quite difficult to explain, but some trends might help).
- We also wanted to understand why the share of women varies so much across departments, and whether performance or something else explains it.
- We shared that this is not only a local trend. We had checked other institutes' student lists by hand and found similar ratios, though comprehensive data is not released.

What we got

1. Admission and enrolment data he showed us

- From the official records (110 admissions in total, excluding BS-MS data), the gender distribution by enrollment is about 1 to 1 in Biology and lower in Chemistry, but "pretty bad" in Maths and Physics. **He stressed that this is not an isolated year.**
- Institute-wide, Biology had 146 women and 134 men (280 in total), a male to female ratio of 0.92. He called this very good, since a ratio near 1 is good and large numbers are bad. Chemistry and Earth Sciences were around 2, and Maths and Physics were much worse.
- Physics has one woman among about 30 faculty. Maths has 1 among 19. He said this small number starts at the student level.

2. The trend over the years

- For BS-MS batches from 14 MS to 21 MS, the male to female ratio in Maths was roughly 5, 8, 12, 4, 4, 5 and 12, an average of about 7. Physics and Earth Sciences are also bad. He called it a clear trend, and said the ratio is very bad in three departments at IISER Kolkata.
- Chemistry is okay, and its ratio is not very different from the BS-MS intake ratio of about 27%. He said the intake is almost always between 25 and 30% and called 27% a good number given the previous years' data.
- For PhD, Biology has more women and the other departments fewer.

3. His summary: it is not about competence

- He said the hypothesis people usually tend to form first is with regard to capability. To test it, he showed us combined CGPA data across batches, which shows that women generally do better than men once admitted. So **capability is not the reason for the low percentage of women students**.
- There is a gender disparity, but it has nothing to do with overall competence. Statistically, women are more competent, with higher average CGPA. They are simply choosing different departments.
- He then told us that he would provide us with the subject-wise SGPA data for the first two years, as it might be a better fit for explaining the differences across departments. We have compiled reports for the same [10], [11] and [17].
- He said it is a small puzzle for the institute too. Earth Sciences (which also has a poor gender ratio) is not a mathematical subject, so it cannot simply be about maths. It aligns more with Chemistry and Biology, yet its ratio is also poor.
- He said the institute has never tried to solve it, mostly because it does not know how to proceed. It would need systematic study from many angles.

4. The question-framing study

- Rohini Godbole, the 2024 convocation speaker, showed an interesting study: the same question, framed differently, gives different marks for boys and girls.
- This shows that the differences arising might be due to a systemic bias in designing tests and educational materials, not due to capacity itself. The study referred to is by Gladys, Furst, Holdsworth and Dastoor, "Gender bias in first-year multiple-choice physics examinations," (Physical Review Physics Education Research)

5. The analysis he recommended that we run

- Undergraduate: look at first- and second-year GPA by gender, since major choice is made after that, and see whether grade performance influences which major students choose.
- PhD: look at the gender ratio by department across years, to see whether some departments are traditionally preferred by women even without bias.
- We told him PhD is slightly better than BS-MS, at about 32% against about 27%. He expects Biology, Chemistry and Earth Sciences to be better than BS-MS, was unsure about Physics, and said Maths numbers are so small that one student changes the picture.

6. Inputs we received

Sir very kindly allowed us to use the anonymised gendered combined CGPA and the subject-wise SGPA data. This data has shaped our further reports and analysis.

7. Why do women drop off later?

- From personal experience: during the PhD, many women face family pressure to marry and start a family. A small group is given freedom and told to marry in good time. A much bigger group marries, has children, and continues studying in parallel.
- He said that after marriage and children, women almost always spend much more time on the family than their male counterparts. He has seen many cases of women dropping their careers after marriage, and very few cases of the reverse. Even when both work, the child goes more to the mother.
- Among faculty, the gap often traces back to the early career years lost to raising children. By the time women return, they have lost the momentum that they had.
- We looked into this aspect as well. Report [7]

8. SC, ST and OBC faculty numbers

- At the end of the meeting we asked him why SC, ST and OBC faculty numbers are so low. He confirmed what we had heard from the DoFA office.
- Until 2019, the roster recommended following the central reservation policy but did not mandate it. If a post reserved for a category was not filled by someone from that category, the institute could fill the vacancy from other categories.
- After 2019 the rule changed: if a suitable candidate from the category is not found, the post has to stay vacant. New faculty recruitment has also been stopped unless special drives are held to hire SC, ST and OBC candidates.
- As a result, he said, the number of SC, ST and OBC faculty has been increasing in institutes across India since 2019.

The next phase

- We have compiled the following three reports: 10, 11 and 17.
- Sir pointed out that women may drop out from higher education due to various reasons including family pressure to get married. We looked into whether the drop actually happens. The report is attached here [7].

Interview 3: Women in STEM at IISER Kolkata

Students, faculty and role models

Dr. Anindita Bhadra [DBS]

Why we went

- We wanted to know why the share of women is low among both students and faculty at IISER Kolkata.
- We shared our faculty data: in DBS there are 9 women to 17 men, and the ratio is worse in other departments. We asked why women are better represented in the biological sciences than in physics and maths, since this is not only a local trend.
- We shared early findings from our survey of female students. Responses are still few, but scholarships and cost, families being worried about sending daughters, and the lack of role models came up. The lack of role models was mentioned by almost every second respondent.
- We asked about her role models and about the challenges she has faced as a woman in STEM.

What we got

1. Why there are fewer female students

- Hostels: feedback from current students about hostel facilities may deter parents of incoming students. This also affects PhD students, who mostly do not have a hostel facility.
- A change in the student mix: IISER Kolkata has historically had many girls from Bengal and Kerala. The Malayalee population has been falling over the years because there are now two IISERs in the south.
- The data since 2013 shows that women faculty are about 13.6 to 14% of faculty at IISER Kolkata and about 25.2 to 28% of students. The share has never gone beyond this range.
- TVM has almost 51% female students and Tirupati about 52%.
- She tested the idea that IISERs with more students from their home state have more girls, since families may prefer a nearby institute. We looked into her suggestion; this analysis was run in the [3] (Section 5, main findings). In brief, the hypothesis did not hold: Kolkata has one of the highest home-state shares but one of the lowest female shares, while Tirupati has quite low home-state students.
- Later, ma'am said that nearby states could also be a factor. Though it seemed intuitively true because of the student distribution by region that we see at IISER Kolkata, we did not have the data of other institutes to test the hypothesis.

2. Why there are fewer female faculty

- IISER Kolkata is located away from the city, and women faculty often have to consider their families.
- If the spouse works elsewhere and there are children, keeping the family together and travelling up and down becomes a problem. Many of her colleagues face this, and it deters candidates who can find jobs together closer to a city.
- Many women faculty, including her, joined early and their husbands are also here, which helped. Now that recruitment has slowed, hiring two people at the same time is not always easy.

3. Why women are better represented in Biology than in Physics and Maths

- She called it a global trend that is more social than anything else, and said it has been discussed in many forums. Even among students, there are more women in the biological sciences overall.
- Girls are not encouraged to study science, and even when they do, they are not encouraged toward the classically "hard" sciences like physics and maths. Biology is somehow expected to be for women. This was true even when she was in college.
- The same divide shows in other fields: more women study medicine than engineering.
- US studies found the divide starts in high school, where girls fear being called nerds. She thought that may not apply in India. Here, families push girls away from the hard sciences, and the notion that maths is for boys gets ingrained in school.
- Families are also uncomfortable with their daughter being the only girl, or a minority, in a class. She called it a multilayered social issue that is very difficult to deal with.

4. Role models

- She agreed that role models matter and said she has worked on this on different platforms. In many schools she has asked students to draw a scientist, and most draw an elderly white man with Einstein-like hair, in a white lab coat, holding a test tube. This starts very early.
- She asked why this should be a deterrent. If you are interested, you should take it up and become another female faculty member. When girls first started going to college there were only male teachers, so it has to start somewhere.
- We told her that 38% of survey responses said the professor's gender does not matter, and about 50% said it matters a lot. She said she understands role models, but that in research what matters is the work, and interesting work should not be gendered. She also said she was not sure this view is universal among students.

5. Her experience as a woman in STEM

- Female students, and also male students, come to her to talk about issues on campus. For the girls, it may matter that she is a woman who can understand their problems. However, she could not be sure of that.
- She faced no problems as a student. After she moved into a job, she said the institute has given her a lot of heartache. She has written about it in an article in India Bioscience, republished by Life of Science, and said we may reference it since it is in the public domain.
- **Anecdotes from the article** (published on our wiki with Dr. Bhadra's consent):

a. "I realized that in order to find stable ground, simply working hard wasn't enough. I was a woman, and I had to handle flirtatious comments, politely reject invitations to dinners, lightly step over compliments on how I dressed…"

b. "He said he agreed that my CV was stronger than some of my colleagues, but he could not give me a regular position due to lack of sanctioned posts. He then told me, *'you are a woman, you have a family, your husband is here and he has a permanent job. Why are you bothered about a career? I will make you a Scientific Officer, and you should be happy. You can teach if your HoD agrees, and even be a co-PI in grants. That way you can also do some research.*'"

- Before academia, she did not face the social issues that we discussed or which tend to discourage women in general.

6. Resources she pointed us to

- The Cogito women advisors series, a set of detailed interviews with women faculty, first at IISER Kolkata and then across the other IISERs. The website is still up. She said we can quote from it with a reference link.

What we learnt

- The reasons for low women students and faculty at IISERs could be due to:

a. Social issues and the traditional leaky pipeline problem. Women not being encouraged to pursue the "hard" sciences

b. Location of IISER Kolkata

c. The two-body problem

d. Though the home-state correlation did not hold [3], she pointed us towards a more likely reason: at each IISER, the percentage of students from neighbouring states could be higher. We, however, have not found any way yet to test the hypothesis. Future iGEM teams are encouraged to do so.

- We also realised that the environment for women in STEM is usually much more challenging than it is for men. Even highly qualified women scientists like Dr. Bhadra have had to face misogyny and comments directed at her by virtue of her being a woman. We realised that the reasons for low faculty count are much more deep-rooted and entrenched than what we had believed earlier.
- We realised that these issues are too large for us to tackle; we could only hope to make a difference via small initiatives to raise awareness.
- After the talk with Dr. Bhadra, we realised the importance of having visible female role models.
- Taking inspiration from #WIISER, a set of detailed interviews with women faculty, first at IISER Kolkata and then across the other IISERs, hosted by Cogito, the in-house science communication magazine, we have planned a women-in-STEM talk series to inspire the women at IISER Kolkata. The details are mentioned in a later section.

Interview 4: Women in STEM at IISER Kolkata

Faculty experience, student numbers and suggestions for the institute

Interviewee: Dr. Malancha Ta (DBS)

Why we went

- We shared our findings so far: more PhD students come from the home state, but we could not find the same pattern at the undergraduate level.
- We asked why the share of women faculty at IISER Kolkata is low, and why the share of women students is low.
- We shared our survey result: around 1 in 2 responses said role models would have made a difference. The sample was small because we had little time.
- We shared the faculty numbers: DBS has 9 women and 17 men, DPS and DMS have 1 each, Chemistry has 5, and DES has 3. We asked why the ratio falls from Biology to Chemistry to Earth Sciences to Physics to Maths.
- We asked whether role models matter, and what she has faced as a woman in STEM.

What we got

1. Why there are few female faculty

- She said it is hard to pinpoint. In her time here, the institute has had four directors, so it is not tied to one person's decisions.
- The most likely reason is the two-body problem. Unless the husband also finds a job in the region, women may not apply or may not accept an offer.
- Kolkata and its surroundings have less industry than cities like Bangalore or Hyderabad. Only AIIMS and IISER are nearby, travel from the city is not easy, and a husband working at an IIT, for example, makes it hard to set up a career.

2. Why there are few female students

- She could not give a reason. Entry is at the national level, so the home state should not matter, except possibly for PhD numbers.
- All undergraduates across the IISERs get hostels, and marriage and spouse-job issues do not apply to them, so she said she simply could not guess.
- She suggested we check the hostel situation at other IISERs, noting that hostels only affect PhD students, since all BS-MS students stay in hostels.

3. Role models

- She does not think a lack of role models explains the low numbers.
- Leadership is where representation is weak. The fact that women make up about 25-27% of faculty in the INIs, but only about 11% of directors and deans, is worrisome. She called these numbers pathetic.
- She said that role models are not as important as women being in positions of power.

4. Where the pipeline changes

- Girls do slightly better than boys in the school boards every year, and in her own lab PhD intake is often more female. So something changes along the way.
- She pointed to the Indian social set-up, where women end up carrying more of the work at home, however much we say equality is maintained.
- Her advice to women students: every day is a fight, so be prepared for it, keep your goals clear and your priority list ready, and you can achieve what you wish. After that, things become easier.

5. Why the ratio falls from Biology to Physics and Maths

- She saw the trend even at school level. In her Delhi school, one of the renowned ones, a class of 43 in the technical-drawing stream had only 3 girls and 40 boys.
- She said she does not think there is any real brain-based reason, since no such distinctions have been found. She suspects it is more about the Indian social condition, and noted that it is a global trend.
- One possibility she raised is that even educated parents may unknowingly steer a girl toward biology when she needs guidance on choosing subjects.

6. Women in decision-making

- She said she thinks women's decisions are more balanced and compassionate, and take into account smaller details that male colleagues may miss, such as in committees and in the student section. This has nothing to do with intelligence.
- Her view was that the improvement may come from the decisions women make, not from girls being inspired by seeing women in power. In other words, the system changes rather than one person being inspired.

7. Her own experience

- She did not feel any disadvantage as a woman through her education. However, she feels it more now that she holds a position. She cannot give a specific example, but there have been times when she felt a male colleague in her place would have been answered differently, sometimes in a condescending tone, from staff or from another faculty member.
- Male friends often say women are being sensitive or overthinking, while women immediately understand. Her advice to younger women is to accept it, move on and do better, so that it gets better for the next generation.

8. Her suggestions and comments on ours

- Scholarship: we suggested one or two tuition waivers per semester for the best-performing women in a batch. She is not in favour of treating women separately, and said she does not even want a Women's Day. She also acknowledged that systemic problems start people far behind the line, so some incentive, not necessarily monetary, would help.
- Targeting is hard: well-off families in the north also push daughters to marry after a BSc, so money is not the only barrier. A merit scholarship can itself persuade a family to let a daughter study, but there is no way to verify who needs it, and a marks-based award may go to someone who does not.
- Safety: she would prefer the institute to provide safe, trusted transport (its own cars or drivers) for women arriving from Kolkata airport at night. She said the institute cannot change everyone around, but it can provide basic safety.
- Hostel: we told her the DoFA says they are speeding up a hostel for female PhD students, especially those from outside the state. She noted it is already being constructed.
- Faculty facilities: the campus already has a well-maintained creche and a campus school. Three or four recently joined women faculty with small children, living away from their spouses, have found these helpful. She suggested more residential facilities to make the campus more liveable.

9. The data discussion

- We showed undergraduate female shares: Kolkata 27.2% and Pune 27.6%, which are among the lowest, and Berhampur much higher at around 43.7%. TVM also has a much better ratio, although all IISERs use the same entry channel.
- She asked whether we have data on how many boys and girls apply through IAT. We have not compiled it. It is not public, so an RTI would be needed. NEET and JEE are both skewed toward boys, and JEE is far more skewed.
- She suggested that parents may simply prefer some IISERs for their daughters, and that other BS-MS programmes in Kolkata, such as at IACS, could be compared. National higher-education enrolment data shows more women than men at undergraduate level in West Bengal, but science enrolment is not split by state. NIRF data is public and can be used without an RTI.

Interview 5: Dr. Anuradha Bhat

Women in STEM at the IISERs

Women faculty at IISER Kolkata

Why we went

- We observed some consistent patterns when we were preparing the reports on gender ratio at IISERs and other INIs. In brief, our report on female representation across the IISERs showed: Tirupati and TVM consistently high, Kolkata and Pune consistently low, Mohali increasing. We had also observed that though women faculty in STEM were a minority in themselves, the numbers tended to be more in biology as compared to disciplines like physics and mathematics. [2], [3], [5], [6], [7]
- We presented our data on female faculty at IISER Kolkata. The number is low in every department, below around one half even in DBS, which has about 9 women to 17 men. Chemistry has 5 women, DPS 1, DMS 1, DES 3, and CDS 2. We asked why the proportion of women is higher in Biology and Chemistry than in Physics and Maths, which is also a global trend. We wanted to understand the trend.
- We wanted to understand why there are fewer women at IISER Kolkata, why the ratio falls across departments, and whether role models matter.

What we got

1. The NIRF and preference correlation (the main insight)

- She helped us see that female representation is almost the inverse of institute ranking and student preference. Pune and Kolkata rank highest and are most preferred, while TVM, Tirupati and Berhampur are lowest on the preference list. The gender ratio runs in roughly the opposite order. This was one of the most direction-changing insights in the entire study.
- Admission is through IAT, a national test with no gender-based reservation. So any gap at admission comes from differences in who gets high ranks. If the top rankers choose Pune and Kolkata, fewer women will qualify for those seats.

2. Why fewer women reach the top ranks: the issue starts earlier

- The women who reach IISER are a filtered group, from families that were supportive, could afford coaching, and backed their education.
- The low ratio may reflect that girls in general receive less family support, and that families invest more in boys' education.
- A possible check is school type. Ma'am also told us that there are reportedly more girls than boys in government schools, which would suggest boys are moved to private schools while girls stay in government ones. This would reinforce the belief that boys' education tends to be better supported.
- Her conclusion was that this is a deep-rooted social issue, not specific to one IISER.

3. Why the ratio falls across departments

- We asked why the proportion of women decreases in the order Biology, Chemistry, Physics, Maths.
- She said the reason is more socio-cultural than anything else. There are fewer women in STEM overall because of social issues, and the drop across departments can also be attributed to similar social issues.
- She pointed to how girls are conditioned from a young age. In a study, she mentioned, teenagers asked to draw an engineer or a scientist mostly drew men, while those asked to draw a teacher or a nurse mostly drew women.
- This socialisation starts long before college and shapes which subjects students feel they belong in and choose.

4. Regional and career-path factors

- Education is highly respected in both West Bengal (the state where IISER Kolkata is located) and the South, and boys and girls are encouraged. But boys are pushed toward engineering or medicine for quick jobs, while girls have more freedom to choose. She pointed to Mohali for this. Kolkata has the same mindset, so she had no firm hypothesis there.
- She suggested looking at the number of Bengalis in the region, since many Bengalis go out for work.

5. Female faculty and PhD students

- Location: IISER Kolkata is remote, which may reduce the number of female faculty.
- Two-body problem: spouses are not offered positions together.
- Campus facilities would help female faculty: a campus school, more residential facilities, and amenities like shops on campus. Some of this is already being done, but more would help.
- The absolute number of faculty at IISER Kolkata is low, so more people need to be hired, including more women. She also said that the low percentages of women at IISER Kolkata may not be a very reliable indicator as the absolute number of faculty at IISER Kolkata is also quite low. She suggested that we look into the absolute number of faculty at all other institutes and compare IISER Kolkata against that. We come to the same analysis later.
- Hostels are not provided for female PhD students, which may be one reason for low PhD numbers.

6. Role models

- She agreed that a female role model makes a big difference. A girl may intuitively like maths, but seeing even one woman in STEM can make her feel more confident.
- Her own role model is Rohini Godbole, who works in a male-dominated field and commands her presence well.
- In 20 years the trend, regarding the participation of fewer women in STEM as compared to men, has not changed.

What we learnt

- The interview with ma'am gave a completely new direction to the way we were approaching the problem. Previously, though we were aware of the social aspects of the low student count (due to the interviews with other female faculty), we had been trying to look at IISER Kolkata as an isolated case study to be studied (the reason why we tried to look into factors like location, home state preference etc.), for the low number of students.
- The interview with ma'am made us realise that probably the gap comes into existence due to a much deeper reason than what we had previously assumed. Institutes which are ranked higher in overall ranking and student preference tend to show lower percentages of women because girls, anyway, are gatekept from facilities like coaching (which would make accessibility to such institutes possible).

What we did next

- As suggested by ma'am, we collected data on female faculty and total faculty across the IISERs, to check whether low faculty numbers explain low female representation: the report is [4.1]

What did we do after receiving the suggestions provided by Dr. Bhat:

Dr. Bhat pointed out that women's share at the IISERs runs almost opposite to institute ranking and student preference: Pune and Kolkata, the most preferred, have the fewest women, and Thiruvananthapuram, Tirupati and Berhampur, the least preferred, have the most. Admission is by rank in one national test with no reservation for women, so a gap at admission has to come from who reaches the high ranks. Dr. De later explained the likely cause: the top of the rank list is dominated by boys, and highly competitive exams need coaching and family investment, which families are often more willing to make for sons. She stressed that this is not about ability. We checked whether the data support the pattern. The full analysis is Report 13 (Rank and Preference).

What we found

- We took women's share among UG five-year students at each of the seven IISERs from the NIRF tables (Report 3) and compared it with each institute's position in the NIRF ranking order, using Spearman rank correlation.
- Women's share rises as an institute sits lower in the order: correlation 0.82 across the seven IISERs (exact permutation $p = 0.03$). Pune and Kolkata, first and second, have 27.6% and 27.2% women; Tirupati and Thiruvananthapuram, fifth and sixth, have about 52%.
- The order the interviews describe as student preference gives a similar result: 0.86 ($p = 0.02$).
- IISc Bangalore, placed first by our own judgement as the most coveted institute, has 16.9% women among UG students. Adding it gives 0.88. We treat it as a reference point, not as extra evidence, because it is not an IISER and its programme is four years.
- Dropping any single institute leaves the correlation between 0.71 and 0.94, so no one institute drives it.
- Women's share among faculty (Report 4) shows no consistent pattern (−0.32 for the seven IISERs, 0.12 with IISc), which fits the view that the gap at admission is a different matter from the gap in hiring.

What this does and does not say

- It is a pattern across seven institutes, **NOT PROOF.** The uncertainty is wide, and rank is tied to an institute's age, size and region.
- Location could produce the same picture. Dr. De noted that families may find the south safer, and Dr. Ta suggested parents may prefer some IISERs for their daughters. This data cannot separate those explanations from rank.
- The mechanism, that fewer women reach the top ranks and why, cannot be tested with public data. It would need applicants, ranks and the institutes chosen, by gender, which are not public; Dr. Ta suggested an RTI request would be needed.
- It says nothing about ability. Women do as well as men once admitted (Reports 10 and 11).

Interview 6: Faculty Perspective on Gender Ratio

Women in STEM at IISER Kolkata: faculty recruitment and student pipeline

Dr. Neelanjana Sengupta, ADoFA (DBS)

Why we went

- We observed some consistent patterns when we were preparing the reports on gender ratio at IISERs and other INIs. In brief, our report on female representation across the IISERs showed: Tirupati and TVM consistently high, Kolkata and Pune consistently low, Mohali increasing. We had also observed that though women faculty in STEM were a minority in themselves, the numbers tended to be more in biology as compared to disciplines like physics and mathematics.
- We presented our data on female faculty at IISER Kolkata. The number is low in every department, below around one half even in DBS, which has about 9 women to 17 men. Chemistry has 5 women, DPS 1, DMS 1, DES 3, and CDS 2. We asked why the proportion of women is higher in Biology and Chemistry than in Physics and Maths, which is also a global trend. We wanted to understand the trend.
- We wanted to understand why there are fewer women at IISER Kolkata, why the ratio falls across departments, and whether role models matter.

What we got

1. Why there are so few female faculty

- She said there are probably several reasons and that more research is needed before any can be confirmed.
- IISERs keep high standards for faculty recruitment and cannot compromise on them. The institute also receives relatively few applications from women.
- She suspects this is the traditional leakage of female talent along the pipeline. Girls do well at school and often outperform boys, but the numbers fall sharply from graduate education onward, and keep declining through postgraduate study and PhD.
- The causes are multifactorial: social expectations, and girls not being encouraged to be very ambitious.
- Family also plays a part. Society expects women to be the ones to compromise, for example in solving the two-body problem, where both spouses need to be in the same place.
- She called it a coupled problem and said she could not prescribe an immediate solution. She added that the faculty numbers we quoted reflect her generation or up to one generation behind her, and she is encouraged by the current generation.

2. What the institute is doing

- There is an unsaid policy that when two applicants are equally good, the woman is selected. We had heard the same from the DoFA.
- She stressed that the institute is actively trying to increase numbers, but it is hard when the pipeline is dry.

3. Why the ratio falls from Biology to Physics and Maths

- The causes lie further back in the pipeline. Women are a minority in both JEE and NEET, but the gap is starker in JEE Mains than in NEET.
- Biology is a more welcoming and broad field, where people from all kinds of disciplines can find a niche. She and Bidisha ma'am studied physics and Koel ma'am studied engineering before moving to biology. She also mentioned a visitor from Emory trained in physics, and a friend at IISc Bangalore with a B.Tech from IIT Kharagpur who now works in molecular biophysics. Chemistry faculty are also often trained in physics.
- She rejected the idea that women lack the talent for maths and physics. In her view, Biology and Chemistry have more diversity within the disciplines themselves, while Physics has yet to break its disciplinary boundaries. For Maths, she suggested asking Koel ma'am.
- Her conclusion was that the start of the pipeline needs reform, and that the society at large (including her generation of parents) carries part of the responsibility.

4. Student numbers

- Our batch has about 75 women out of 299 students, roughly 25%, which is on the lower side. She took some comfort that it is still above the ratio which was there when she was pursuing her education. She said that things are slowly getting better.
- She would like to see a minimum of 40 to 45 percent of women in all fields, and believes this would lift numbers at the end of the pipeline. It would not happen overnight and must be nurtured early.
- She noted that the current generation can help both boys and girls to grow with equal ambition and equal opportunity, and said everyone can do their bit.

5. Role models

- She agreed that role models matter, and said she had raised the same point in an earlier student-club interview. Growing up she knew of Marie Curie, but all her MSc professors except one were men, and she had never met a woman scientist in person before.
- She said it is natural for a woman to not want to sacrifice her personal life for a career. For this, proper support must be provided. She hopes a few faculty like her can partly fill the role-model gap.
- We told her that female students who want to take Physics are put off because there is only one female professor in the department. She said she would raise this with the DoFA and that it is high time the Physics department grew.
- She suggested we also interview the HODs, especially the Physics HOD.

6. Challenges faced by women, and tokenism

- Conferences now fund and invite based on female representation. However, she said that ***it is insulting when a host says she was invited because they needed women, and wishes male colleagues were more sensitive***.

- She also finds it patronising to be told in a committee that she is there as the female representative. She asked us to quote her on this, without naming anyone, and said it should not be done. This form of **tokenism and patronising attitude must stop**. Respect is not something that can be manufactured.

What we learnt

- The reasons for the low ratio of women might be:

a. Location of IISER Kolkata in a remote area

b. Social Issues: the leaky pipeline in STEM due to social expectations, and girls not being primed to be very ambitious.

c. The two-body problem and the social expectation that women should be sacrificing their careers rather than the men

d. Lack of role models

e. Patronising attitudes and tokenism towards female scientists who do make it to the end

f. Biology probably has a better ratio as the field of biology is itself broad and welcoming towards all disciplines

Interview 7: Women in STEM at IISER Kolkata

Faculty recruitment, women in decision-making and student enrolment

Dr. Koel Das (HOD, Department of Mathematics and Statistics, IISER Kolkata)

Why we went

- From the DoFA office we collected faculty numbers. We also had the student numbers from NIRF. IISER Kolkata is among the lowest in both, compared with the other IISERs and with the national average. The DoFA office said women do not apply much in departments like physics and maths, so there are very few applications to consider.
- The female faculty count by department is low everywhere: Biology has 9 women to 17 men, Chemistry 5, Earth Sciences 3, and Maths and Physics 1 each.
- In our survey of students, one of the reasons for fewer female students was the lack of female faculty and role models in many fields.
- We asked her why female faculty numbers are low, whether students approach her about this, whether female role models make a difference, and what she has faced that might also stop people from applying to IISER Kolkata as faculty or students.
- We also asked why the ratio of female faculty tends to be more concentrated in the senior ranks rather than the lower ranks at IISER Kolkata (which is anomalous compared with the national trend).

What we got

1. Why recruitment of women faculty has been low

- She confirmed that the issue is two-fold: the lack of female faculty, which is a global and local problem, and the enrollment of female students.
- Most women faculty at IISER Kolkata are senior, because the institute was new and many faculty joined around 2010 to 2012. Since most of the women faculty joined during that period itself, they have generally succeeded in making a mark and have been promoted to higher positions. At that time, the female to male ratio was not as bad at IISER Kolkata, as compared to the current day.
- Faculty recruitment since then has been very slow compared to other IISERs such as Pune. There was no general recruitment after around 2018 or 2019, until the current all-open call, and only two special recruitments. She put this down to administrative reasons and did not go into detail.
- She noted that Pune and other IISERs have recruited far more aggressively, so IISER Kolkata has a gap.

2. Location and the two-body problem

- IISER Kolkata is in a semi-urban location, so it matters whether the spouse can find work nearby. Apart from IISER and AIIMS, there are only a few options, unless the spouse is also in academia.
- Living in Kolkata and commuting is hard, especially for experimental work that needs long hours in the lab.

- When she joined, IISER Kolkata judged each applicant independently, even if the spouse was also faculty. Many older institutes would not even consider both spouses. She said we should incentivise this more and make it clear that we are open to hiring spouses.

3. Policy on hiring women

- There is no reservation for women. But if two candidates are at par, women are generally preferred. The DoFA office told us the same.
- The DoFA office also said they do not get many applications from women, and the ones they get are usually in fields where faculty already exist. In the current search in physics, they could not find a suitable female candidate.

4. Why women are fewer in Physics and Maths

- The numbers are highest in Biology, then Chemistry, then Physics, then Maths. It has always been like this, and it is the same globally. Theoretical physics and maths have the fewest women.
- Ma'am told us that studies in the West point to a social cause: girls are discouraged from maths and competition because being called a nerd is looked down on. She said this is not the case in India, where being good at maths earns respect regardless of gender. She has not seen Indian studies on why women faculty and PhD students drop.
- She was clear that this is speculation. She thinks it relates to the long path of a PhD, a postdoc and then a faculty job, and to societal pressure to settle and get married.
- In a patriarchal society, a woman's career is often guided by where her husband works rather than by her own factors.
- A positive sign: in the last couple of years the number of female PhD students in maths has risen from almost none to a better, though still not healthy, number. She said a minimum group of women is needed, because a lone woman has a good chance of quitting maths or theoretical physics.
- She said the same trend appears among students. DBS has many women at undergraduate and PhD level, while the other streams do not.
- From her memory of US education research, the segregation starts in high school. She felt that here the reason is mainly societal, though she could not point to another cause.

5. Women in decision-making roles

- She has not faced problems inside the institute. However, outside the institute, the problems came from some older men while she was presenting for grants. At one grant defence soon after she joined, the chairperson addressed her in a condescending way as "my dear girl". Though she later got the grant, the incident was negative to say the least.
- She said one way to address the role-model problem is to bring more women into decision-making: as deans, directors and members of selection committees. At IISER Kolkata all the deans are men.
- We mentioned that women constitute **around 25% of the faculty overall** at INIs. **Across the history of all Institutes of National Importance (which includes all IITs and IIMs), only four women have ever served as institutional Directors**. Moreover, a 2025 survey of 1,352 Indian higher-education institutions found 11.2% headed by women. At Institutes of National Importance it was 14 of 164

(about 8.5%), and no IIT was led by a woman. Women are 31.3% of professors in India, so the pool for these posts is larger than the posts they hold.

- Dr. Das also mentioned that to date, she can recall **only one female Dean** at IISER Kolkata.
- She argued that this should be enforced even if women are not great administrators at first, since they will become good with time, as male administrators did. She said Indian science decision-making is dominated by older men, which is the root of the problem.

6. Why the undergraduate share is lower than the PhD share

- We noted that the share of women at undergraduate level is lower than at PhD level. She thinks location plays a part, along with the lack of hostels.
- For undergraduates this is probably less of an issue since they get hostels. For female PhD students it is a big gap. Parents of students from other states often worry, especially when their daughter is leaving home for the first time.
- She gave the example of a current PhD student whom she had to assure of a hostel. The student first stayed in a dorm room because there was no space.

7. Her suggestions

- Incentives for women: a proposal to give awards, scholarships or tuition waivers to top-performing women in the BS-MS and PhD programmes was made at one point but never implemented. She said it does not need to be many, even one or two would do, and that we should offer something other institutes are not offering.
- Hostels for female PhD students.
- A clear and visible policy of hiring spouses to solve the two-body problem.
- More women in decision-making roles.

8. Role models

- Personally, though role models did not shape her career (she comes from an engineering background, where the ratio is extremely skewed), when we told her that many female students in our survey said it matters a great deal if their professors are women, she said she could see why.
- She also mentioned that sometimes, she could see that perhaps female PhD students are more comfortable talking to her about issues than to her male colleagues, even though they are approachable.
- She said she sees no disadvantage in having more women in positions of power, only possible benefits, so there is no reason not to.

What we learnt

- We realised that although fewer women in STEM and in faculty positions is a global issue, the anomalous numbers at IISER Kolkata might be partly due to the slowing in the hiring process after the initial recruitments.

- Moreover, the high concentration of the women faculty at IISER Kolkata could be due to the fact that most of them joined the institute during the early years (when there was a lot of recruitment ongoing). Since then, a lot of time has passed. Everyone tends to be promoted over time. She also made us realise that not just women, but the total faculty pool at IISER Kolkata leans towards senior ranks. This was also brought up by the DoFA.
- The location of IISER Kolkata might be a deterrent to female faculty, especially keeping in mind that in a patriarchal society, it is usually a woman who is asked to compromise when solving the two-body problem.
- We realised that the differences in women across departments could be due to the socialisation process and societal norms.
- She recommended that more women in positions of administrative power (like directors and deans) could significantly help alleviate the current status of women in STEM.
- She also recommended that finding a solution for the two-body problem (for example, by visibly being open to hiring spouses) might improve the ratio of the female faculty.
- The low number of PhD students could be due to the lack of hostel facilities.
- Incentivising the women who perform well (via awards, scholarships and tuition fee waivers) and increasing the number of women in positions of administration, could alleviate the situation.
- She also said that visible role models could help inspire young women and girls.

Interview 8: Women in STEM at IISER Kolkata

Student numbers, the rank and preference correlation, and the leaky pipeline

Dr. Rumi De (DPS)

Why we went

- We shared our briefs and reports. Across India, women are about 49 to 52% of undergraduate students. In the INIs the average is lower; however, IISER Kolkata sits even below that at about 27.2% and has stayed between 26 and 28% for a long time, with Pune at 27.6%.
- PhD percentages are marginally better, but the share of women faculty is very low at about 14%. TVM and Tirupati have much better women's representation.
- We asked why the student share and faculty shares are so low at IISER Kolkata. We also shared our home-state data, our rank and preference correlation, and the results of our survey.
- We also wanted her recommendations on the survey we had done for the women in nearby villages.

What we got

1. Why fewer women choose IISER Kolkata

- Allocation is central, so it depends on where students want to go. Pune and Kolkata are the most preferred, yet both have few women, so she said it is not about IISER Kolkata itself.
- She thinks parents' concern matters a lot for daughters. BS-MS is residential, five years long, and starts at a young age right after Class 12, so parents have to be confident about safety.
- She felt that Tirupati and TVM do better because the south is seen as safer, literacy and awareness are higher, and parents send daughters and sons for higher studies equally. Parents also prefer a local option they can reach, and southern students can choose the southern IISERs. This is her perception of the data.
- Many Bengali girls join IISER Kolkata. She said parents elsewhere may hesitate to send daughters over a longer distance.

2. Home state and nearby states

- We showed our home-state data. Kolkata and TVM both have a large share of home-state students, and Tirupati has a small share but a high share of women. So home state alone does not explain it.
- She suggested including nearby states, since many students choose the closest IISER (for example, students from Odisha, Assam and the northeast choosing Kolkata) and parents prefer somewhere they can reach quickly. This is a suggestion that we received from Dr. Anindita Bhadra as well.
- Our NIRF data only has home state, not nearby states, so this analysis may not be possible with the data we have.

3. The rank and preference correlation

- We showed that the most preferred IISERs, in the usual order Pune, Kolkata, Bhopal, Mohali, then TVM, Tirupati and Berhampur, have the worst gender ratios. We found the same pattern for the IITs.
- She explained it: admission is rank-wise, and the top rankers get their preferred institute. The top of the rank list is dominated by boys, and there is no reservation. So the data of girls not choosing Pune or Kolkata may, in fact, depend on their ranks in IAT. **She stressed that this is not about ability.**
- The cause is **guidance and resources.** Highly competitive exams need coaching and support, and parents are often more willing to invest in sons because they expect them to earn and look after them, while daughters are expected to marry. In ordinary colleges, the undergraduate and postgraduate ratio is about 50%, because the competition is less cutthroat.
- We told her that when we first compiled this, we hesitated to present it because it seemed to show girls falling behind. She said that data is data, and that the reason is probably social: socio-economic and parental background, coaching and a supportive environment.
- She called this a symptom of deep-rooted issues with a historical background. It reflects how parents raised children 20 years ago, and it is changing, but slowly. She said in some northern states girls often drop out of school and marry early.

4. Performance once they are here

- We got batch-wise CGPA data from Dr. Rangeet Bhattacharyya. Women start slightly behind, then catch up and do equally well or better. In the COVID period, women did worse in the first semesters and then caught up. We speculated this was due to household chores, but it is only speculation.
- Subject-wise SGPA for the first two semesters: in Physics, women do marginally better in some years only. In Chemistry they are the same or better. In Biology they consistently do better, and in Earth Sciences usually better. Maths is the exception: apart from two years, women were equal or lower. Batches are small, so averages can swing.
- Her reading: once on an equal platform and away from home, women do as well or better, so the gap is not capability. Maths is hard for everyone, and she tells us that students tend to take it only if they are passionate. Women who choose it and join the maths department do very well. The gap in performance subject-wise is most probably due to the social conditioning that happens since a young age wherein girls are discouraged from pursuing certain subjects.
- We discussed that any gender-based differences on campus have come from peers, not from professors, and that classes and resources are equal for everyone.

5. Why there are fewer female faculty

- The entry pool is smaller. If women are about 27% of the pool, then the same success rate yields far fewer women, and she said that she found it difficult to understand why faculty is 14% when entry level is 27%. She said that some of this comes from small intakes in some subjects. In one year, Physics had only two women out of 50 students.
- Ma'am further informed us that the academic path is long: Masters, PhD, and often more than one postdoc. This coincides with the stage of life when family responsibility comes in. Childbirth and early childcare mean a break, and after a break it is very hard to come back.

- She said there is no reservation for women, and no policy that credits a career break in selection. Peers have more publications by then, and selection committees do not make up for the break. She thinks such a policy should exist.
- Even with a supportive partner in academia, the woman often compromises, taking a college or other softer position instead of a PI position. Women also think twice about when to start a family in ways that men often do not.
- A ray of hope: a government grant now supports women who have taken a break, with an independent position at an institute. Grants are about three years, which she thinks is too short. She wants longer ones, and said you can apply for a second grant. She said the government and private sector should both provide this.
- She also said motivation has to come from women themselves, since some settle for easier options. Everyone, boy or girl, should have the choice to join academia, industry or elsewhere.

6. Equity and role models

- She believes in giving a jump start so that people reach an equal platform and can then excel on their own. Without it they lag behind.
- Stereotypes like girls cannot do maths start early and should be challenged by everyone, including relatives. She said it is about opportunity and encouragement, not intelligence.
- Women role models help: a woman teacher, a cousin or a sibling doing a PhD can make a girl think she can do it too.

7. Her own experience

- She said students need a supportive environment, space, peace of mind, and time to fail and learn.
- Students, both boys and girls, approach her about majors and summer research. She is a member of the grievance cell. She said some issues are easier to discuss with a woman, and that it helps to have a senior person to talk to. She encouraged students to talk to a senior, faculty member, parents or the Mind Care and Wellness Centre (MCWC) of IISER Kolkata when required.

Interview 9: Follow-up to the interview with Dr. Rumi De (Interview 8): local NGOs and skills training

iGEM IISER Kolkata 2026 · Inclusivity · Follow-up to Interview 8, the interview with Dr. Rumi De

Why we went

Our survey [12] on the women in the neighbouring villages showed some trends that we wanted to address. We had noticed that though the literacy was low, they were very amenable to pursuing further education, especially hands-on skills like sewing, etc. We wanted to discuss if it's possible to find a way to do the same.

What she suggested

Dr. Rumi De suggested that we approach local NGOs to discuss providing training in practical skills such as sewing and computer use, and, where it fits, adult literacy classes, for the women in the settlements next to our campus.

Why this fits what we found

- **The women were interested.** In our survey, a significant proportion of the women were open to learning, and some were enthusiastic. Of the 77 who answered, 43 said they would like to study or learn something now, and a practical skill such as sewing or computers was the most common thing they named (about 20 women).
- **A skill can bring a little income of their own.** Many of the women told us that cost and poverty were why they could not continue their education (55% named it as a reason). A small income of their own could ease that. Most of them also want their children, especially girls, to study, so extra money would be useful for fees and books.
- **Being able to sign their own names.** During the survey we observed that being able to sign their own name was something women of all ages wanted. Adult literacy classes could help with this.

What we did with it

- These suggestions require long and sustained efforts; we are still looking into whether it will be possible to implement them. We shall be approaching Ek Pehal and the local bodies once we have a plan in mind.
- (planned) When we repeat and extend the survey in later phases with Ek Pehal, we shall be directly asking which skills women would like to learn, and how much time they can give.

Limits

The survey is a small convenience sample, so these numbers show what the women we met said, not what all women in the settlements would say. We have not tested whether a training course would raise anyone's income. That is a question for the NGOs and for a later team.

Part B: Fisherfolk and fish ecology

Interview 10: Interviews with the fishing folk

Part 1

Ground-level view of the decline in fish and identifying sex

Interviewee: Mohadep Dutta

Experience: Has been catching fish since he was 7 years old

Why we went

- To hear directly from a fisherman how the catch has changed, how the rules affect his work, and whether fishermen can tell male and female fish apart.

What we got

1. Telling male and female fish apart

- For some fish, fishermen can tell the sexes apart. In prawn (golda chingri), males have bigger legs and females have smaller legs.
- For elish (hilsa), they cannot tell the difference.
- He could not say much about whether the ratio of males to females has changed over time. It is not something they notice due to the lack of visible markers.

2. The decline in fish

- The quantity of fish has gone down severely. Nowadays you do not get any fish in the river.
- What he saw when he was 7 years old does not compare with now. Earlier there were fish of all sizes. Now there is nearly nothing, neither big nor small fish.
- Hardly anyone catches anything now. Everyone is sitting idle, because the water is not good.

3. Other difficulties in fishing

- People cut the river banks and put more sand into the river.
- Big ships cut their nets.

4. Support and who to tell

- Nobody helps them much.
- They do not know the reason for the decrease and say it is the Ganga's matter.
- He asked who they would tell about it. If they do raise it, they are asked why they catch small fish.

What we learnt

- There were no reliable, visible markers to differentiate the male fish from the female. We spoke to Dr. Anuradha Bhat next, an expert in fish ecology, to know of any markers other than vitellogenin.
- The number of fish has gone down drastically in the Ganga. This has seriously been hampering their livelihoods.
- There is no one they can complain to regarding this; whenever they try to raise the matter, they're told not to catch the small fish. However, that alone is not enough to sustain their livelihoods.
- There is a gap between the people framing the laws and the people being expected to follow the regulations.

Interview 11: Interviews with the fisherfolk

Part 2

Livelihood, pollution, regulation and changes over a lifetime

Interviewees: Gobinda Shikdar and Sukuma Sarkar

Experience: Have been fishing since 2011

Why we went

- To understand how fishermen's livelihoods have changed, how they see the government's rules on catching small fish, and whether they can tell male and female hilsa apart.

Note: these are our notes taken from the conversation (in Bengali), with translations and minor language cleanup. The sections on food, health and the past reflect the interviewees' own views and have not been verified.

What we got

1. Fishing as a way of life

- Their ancestors were connected to fish, and their whole lives are connected to fish.
- If there is a problem with the quantity of fish or with seasonal changes, they do other work, but their life depends on fish and always has.

2. Why the fish have decreased

- The Ganga does not have as much current as before, and the water is polluted, so there are no fish.
- Water levels in the Ganga have fallen. The river has become polluted, so fish do not come from the sea anymore. They said this several times.
- They put the pollution down to drainage water and garbage.
- Before, they could drink the water and cook with it. Now they cannot, because there are harmful gases and it smells. The Ganga water has to be filtered to drink or use.
- Sometimes they go towards the sea in the hope of a bigger catch, but ocean fish have decreased too.

3. Livelihood and income

- Catching fish no longer gives them a livelihood or lets them support a family. That is why everybody in the family has to work here and there.
- They work on daily wages that depend on the amount of fish caught. Some days they get a lot, but mostly it is very little, and some days there is nothing in the net. They go on because it is their duty.
- They fish from night to morning, for prawn for example, and manage to earn only about 200 rupees. They do not have a business of their own.

4. Government rules on small fish

- They mostly catch hilsa (elish). They cannot easily tell males and females apart, but they also said they pay no attention to this, since all that matters to them is getting enough catch.
- There are multiple government laws saying that small fish should not be caught. They asked how they can follow them: if they did, they would not be able to fill their stomachs.
- There are no big fish in the rivers, and even in the sea they catch small fish. If they gave the fish time to grow up, they would have no food. If they followed everything the government says, there is no way they could support their families.
- They said neither they nor the government can release medicines or chemicals into the river for the fish, because of the rapid currents of the Ganga. They also mentioned that big fish eat small fish.

5. Their wider views

These are the interviewees' own recollections and opinions, recorded as they told us. We have not verified them.

- Development and pollution: earlier there was no such development, and they used Ganga water directly from the tubewell. Now the population and advancements have increased and the water has become polluted. Earlier they carried their own bags to the market, but now every shop gives a plastic bag.
- Land and food: they said food is now polluted too. Cultivable land is being overused and divided for housing because of the growing population, and fertilisers are used, with three rounds of crops on the same land. They said earlier food was less in quantity but better in quality, and that there was no poultry chicken.
- Fish: they said they never saw or ate the colourful foreign fish earlier, and that the market now has fish like Rui from Andhra. They said the type of hilsa they ate with their grandparents cannot even be seen now.
- Cost of living: they said that now people have money but no food, while earlier there was little money but there was food. They recalled rice at 1.5 to 1.75 rupees per kg and daily wages of 1 to 2.5 rupees, and government jobs paying 300 to 500, with a good life. Now people get 8,000 rupees but that quality of life is not there. In their words, 2 rupees back then is equal to about 500 rupees now.
- Health and age: they asked whether we would be able to live to 65 or 70 like them. They said they cannot give their bodies rest because it would get worse, and that they sit on the boat in natural air. They said people now care only about taste and that this leads to many ailments, and that people start feeling pain at around 45.
- Some of what they said about food and vaccines is hearsay and has been excluded by us.
- Education: they said that earlier people were less educated but used their brains, and that now people are more educated but there is no peace.

What we learnt

- The number of fish has gone down drastically in the Ganga due to reasons like pollution and change in the water level. This has seriously been hampering their livelihoods.
- There is no one they can complain to regarding this; whenever they try to raise the matter, they're told not to catch the small fish. However, that alone is not enough to sustain their livelihoods.
- There is a gap between the people framing the laws and the people being expected to follow the regulations.
- The Ganga currents are too strong; no solution can be directly released into the Ganga. This made us realise that our solution would have to be employed inside a WWTP and not released directly into the river or any uncontained environment.
- There is an influx of foreign, non-local fish in the market. We recognise that this might be an issue. Therefore, when we suggest a list of recommendations, we suggest rearing local species of fish rather than non-native species.

Interview 12

Dr. Anuradha Bhat

Professor, IISER Kolkata

Fish ecology

Why we went

- In our fishermen interviews, we found that the government asks fishermen to catch only large fish and avoid juveniles, to protect ecological balance. But we came to know from the interviews that we took with the fishermen, that large fish alone cannot sustain their livelihoods, so they also catch small fish and break the rules.
- This showed a gap between policy, which is based on ecology, and ground reality, which is based on livelihood. We wanted to know how that gap could be addressed.
- The fishermen also told us they could not tell male and female fish apart, especially hilsa. Since vitellogenin is only a chemical signal, we wanted to know whether there is a visible, practical marker.

What we got

On the policy gap

- The government handles pollution the same way. Penalties and regulations exist, but companies often do not follow them, and there is no proper mechanism to keep this in check.
- As a result, enforcement tends to fall on local people, who are easier to fine than industry.
- Her view was that both sides need action: industry must be regulated, and fishermen must be supported.
- She suggested increasing inland freshwater aquaculture (pisciculture) instead of relying on river water, which may itself be polluted. These systems already exist, but their number can be increased. She told us to look into the possibilities of the same.
- These systems should rear local species, not exotic ones, because any leakage into the natural environment could disrupt the ecological balance.
- She also described aquaculture farms in detail. The gap between fishermen and government as such was not her line of work.

On identifying sex

- Females change physically and morphologically during the breeding season. The abdomen becomes swollen and soft because of the eggs.
- This swollen belly is a visible marker fishermen can use to identify females, especially in hilsa. At other times of the year, males and females look the same.
- She also suspects that poor water quality may affect the reproductive process.

What we did next

- Wrote a mail to the Prime Minister's Office describing the policy gap and recommending our solutions.
- Planned to speak to other experts about the issue: Dr. Devajit Basumatari.
- Prepared two brochures (in English and in Bengali) for fishermen explaining how to identify sex using the breeding-season marker [19.1][19.2].